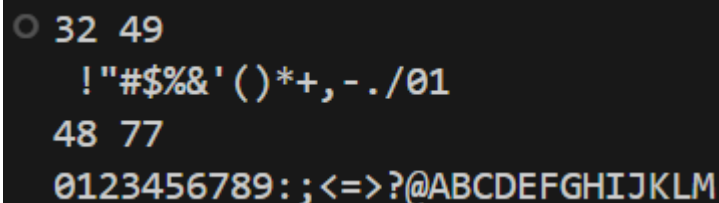


编程练习一

```
1  #include <iostream>
2
3  using namespace std;
4
5  int a,b;
6
7  int main()
8  {
9      while(cin>>a>>b){
10         for(auto i = a;i <= b;i ++){
11             cout<<char(i);
12             cout<<endl;
13         }
14         return 0;
15     }
```

运行结果



```
○ 32 49
   !"#$%&'()*+,-./01
   48 77
   0123456789:;<=>?@ABCDEFGHIJKLM
```

编程练习二

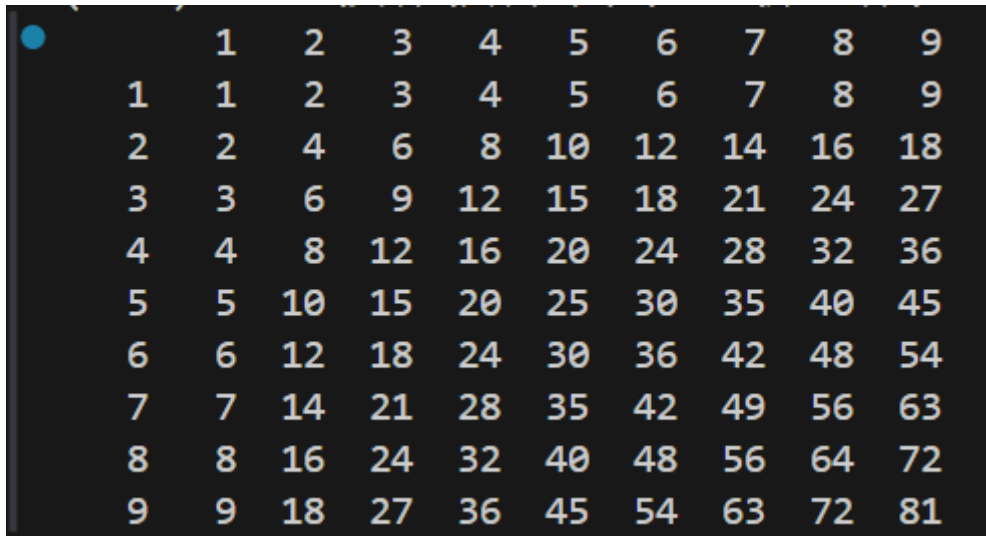
```
1  #include <iostream>
2  #include <vector>
3  #include <iomanip>
4
5  using namespace std;
6
7  vector<int> a{0,1,2,3,4,5,6,7,8,9};
8
9  template<typename T>
10 void print(const T& value) {
11     cout << setw(4) << value;
12 }
13
14 int main()
15 {
16     for(const auto i : a){
17         for(int j : a)
18             {
19                 if(i == 0 && j == 0) print("");
20                 else
```

```

21         {
22             auto result = i*j;
23             if(result == 0) print(((i==0) ? j : i));
24             else print(result);
25         }
26     }
27     cout<<endl;
28 }
29 return 0;
30 }

```

运行结果



	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	10	12	14	16	18
3	3	6	9	12	15	18	21	24	27
4	4	8	12	16	20	24	28	32	36
5	5	10	15	20	25	30	35	40	45
6	6	12	18	24	30	36	42	48	54
7	7	14	21	28	35	42	49	56	63
8	8	16	24	32	40	48	56	64	72
9	9	18	27	36	45	54	63	72	81

编程练习三

```

1  #include <bits/stdc++.h>
2
3  using namespace std;
4
5  vector<string> s{"red","yellow","blue","white","black"},result(3);
6
7  set<vector<string>> res;
8  vector<vector<string> > res1;
9
10 int vit[6] = {0};
11
12 void print1()
13 {
14     int ans = 0;
15     cout<<"There are "<<res1.size()<<" options to take balls out.\n";
16     cout<<endl;
17     for(auto x : res1)
18     {
19         cout<<setw(8)<<ans;
20         for(auto i : x)
21         {

```

```

22         cout<<setw(8)<<i;
23     }
24     cout<<endl;
25 }
26 res1.clear();
27 }
28
29
30 void print2()
31 {
32     int ans = 0;
33     cout<<"There are "<<res.size()<<" options to take balls out.\n";
34     cout<<endl;
35     for(auto x : res)
36     {
37         cout<<setw(8)<<ans;
38         for(auto i : x)
39         {
40             cout<<setw(8)<<i;
41         }
42         cout<<endl;
43     }
44     res.clear();
45 }
46
47 void solve1(int k)
48 {
49     if(k >= 3)
50     {
51         res1.push_back(result);
52         k = 0;
53         return;
54     }
55
56     for(int i = 0; i < 5; i++)
57     {
58         if(!vit[i])
59         {
60             vit[i] = 1;
61             result[k] = s[i];
62             solve1(k+1);
63             vit[i] = 0;
64         }
65     }
66 }
67
68 void solve2(int k)
69 {
70     if(k >= 3)
71     {
72         vector<string> tmp = result;
73         sort(tmp.begin(), tmp.end());

```

```
74         res.insert(tmp);
75         return;
76     }
77
78     for(int i = 0; i < 5; i++)
79     {
80         if(!vit[i])
81         {
82             vit[i] = 1;
83             result[k] = s[i];
84             solve2(k+1);
85             vit[i] = 0;
86         }
87     }
88 }
89
90 int main()
91 {
92     solve1(0);
93     print1();
94     cout<<"-----\n";
95     solve2(0);
96     print2();
97     return 0;
98 }
```

运行结果

● There are 60 options to take balls out.

1	red	yellow	blue
2	red	yellow	white
3	red	yellow	black
4	red	blue	yellow
5	red	blue	white
6	red	blue	black
7	red	white	yellow
8	red	white	blue
9	red	white	black
10	red	black	yellow
11	red	black	blue
12	red	black	white
13	yellow	red	blue
14	yellow	red	white
15	yellow	red	black
16	yellow	blue	red
17	yellow	blue	white
18	yellow	blue	black
19	yellow	white	red
20	yellow	white	blue
21	yellow	white	black
22	yellow	black	red
23	yellow	black	blue
24	yellow	black	white
25	blue	red	yellow
26	blue	red	white
27	blue	red	black
28	blue	yellow	red
29	blue	yellow	white
30	blue	yellow	black
31	blue	white	red
32	blue	white	yellow
33	blue	white	black
34	blue	black	red
35	blue	black	yellow
36	blue	black	white
37	white	red	yellow
38	white	red	blue
39	white	red	black
40	white	yellow	red
41	white	yellow	blue
42	white	yellow	black
43	white	blue	red

```
43  white  blue   red
44  white  blue  yellow
45  white  blue  black
46  white  black  red
47  white  black  yellow
48  white  black  blue
49  black   red  yellow
50  black   red  blue
51  black   red  white
52  black  yellow  red
53  black  yellow  blue
54  black  yellow  white
55  black   blue   red
56  black   blue  yellow
57  black   blue  white
58  black  white   red
59  black  white  yellow
60  black  white  blue
```

There are 10 options to take balls out.

```
1  black  blue   red
2  black  blue  white
3  black  blue  yellow
4  black   red  white
5  black   red  yellow
6  black  white  yellow
7   blue   red  white
8   blue   red  yellow
9   blue  white  yellow
10  red    white  yellow
```